

# affirmoTM tutorial 4: running the model

What `tm_run()` does, and the steps of a run one at a time

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## Table of contents

|          |                                          |          |
|----------|------------------------------------------|----------|
| <b>1</b> | <b>What <code>tm_run()</code> does</b>   | <b>1</b> |
| 1.1      | Run time                                 | 2        |
| 1.2      | The same seed, the same results          | 2        |
| 1.3      | Outputs                                  | 2        |
| <b>2</b> | <b>The steps of a run, one at a time</b> | <b>2</b> |

This tutorial starts from a fresh analysis folder with the example input files (see tutorial 1):

```
library(affirmoTM)
```

```
tm_example_inputs()
```

## 1 What `tm_run()` does

`tm_run()` runs the whole model with the files in `inputs/`:

1. reads and checks every input file and setting; any problem stops it before anything is run or written;
2. lists the death states it found in the matrices;
3. runs the model `n_runs` times, run `r` with seed `seed + r - 1`;
4. averages the runs, compares the arms, and writes the results to `outputs/`.

While it runs, it reports the death states and its progress (here with 20 runs):

```
settings <- tm_read_settings()
settings$n_runs <- 20
results <- tm_run(settings)
```

Death states: CurrentDeath, AB\_Death, AHS\_Death, AIS\_Death, AHS\_AB\_Death, AIS\_AB\_Death.

Run 10 of 20 done.

Run 20 of 20 done.

Check that no living state is among the death states.

## 1.1 Run time

The run time grows with the number of runs and the number of patients:

```
system.time(tm_run(settings))
```

```
   user  system elapsed  
27.160   0.209   6.817
```

## 1.2 The same seed, the same results

```
again <- tm_run(settings)  
identical(results$by_run, again$by_run)
```

```
[1] TRUE
```

## 1.3 Outputs

Every call writes, and overwrites, the files in `outputs/`:

```
list.files("outputs")
```

```
[1] "ce_plane.png"      "comparison_by_run.csv" "comparison.csv"  
[4] "summary_by_run.csv" "summary.csv"
```

Tutorial 5 explains the results.

## 2 The steps of a run, one at a time

`tm_run()` calls the functions below, once per run. They can also be used one at a time, for example to look at one run in detail. Each takes the `settings` list. Tutorial 6 follows a run step by step in more detail.

**Read the inputs.** Each reader checks its file:

```
matrices <- tm_read_matrices(settings)  
cohort <- tm_read_cohort(settings, matrices)  
entry_states <- tm_read_entry_states(settings, matrices)  
costs <- tm_read_costs(settings, matrices)  
utilities <- tm_read_utilities(settings, matrices)  
intervention <- tm_read_intervention_cost(settings)  
death <- matrices$usual_care$death
```

**Entry states.** `tm_assign_entry_states()` draws each patient's state in their entry year. Both arms start from these states:

```
patients <- tm_assign_entry_states(cohort, entry_states, settings, seed = settings$seed)  
patients[, .N, keyby = entry_state]
```

```
Key: <entry_state>
  entry_state      N
  <char> <int>
1:      AB       17
2:      AHS      16
3:     AHS_AB      1
4:      AIS      64
5: CurrentDeath  420
6:     NoEvent  4318
7:      PB       43
8:     PB_AIS      2
9:      PHS      28
10:     PIS      89
11:     PS_AB      2
```

**Simulation.** `tm_simulate()` moves each patient through both arms, one year at a time, until they die or the horizon ends. One row per patient, arm and year:

```
sim <- tm_simulate(patients, matrices, settings, seed = settings$seed)
sim[id == 1]
```

|     | id    | arm        | cycle | age   | agegrp | sex    | mm_cluster       | state   |
|-----|-------|------------|-------|-------|--------|--------|------------------|---------|
|     | <int> | <char>     | <int> | <int> | <char> | <char> | <char>           | <char>  |
| 1:  | 1     | usual_care | 0     | 65    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 2:  | 1     | usual_care | 1     | 66    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 3:  | 1     | usual_care | 2     | 67    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 4:  | 1     | usual_care | 3     | 68    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 5:  | 1     | usual_care | 4     | 69    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 6:  | 1     | usual_care | 5     | 70    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 7:  | 1     | usual_care | 6     | 71    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 8:  | 1     | usual_care | 7     | 72    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 9:  | 1     | usual_care | 8     | 73    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 10: | 1     | usual_care | 9     | 74    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 11: | 1     | iabc       | 0     | 65    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 12: | 1     | iabc       | 1     | 66    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 13: | 1     | iabc       | 2     | 67    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 14: | 1     | iabc       | 3     | 68    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 15: | 1     | iabc       | 4     | 69    | 65-69  | men    | Neuropsychiatric | NoEvent |
| 16: | 1     | iabc       | 5     | 70    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 17: | 1     | iabc       | 6     | 71    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 18: | 1     | iabc       | 7     | 72    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 19: | 1     | iabc       | 8     | 73    | 70-74  | men    | Neuropsychiatric | NoEvent |
| 20: | 1     | iabc       | 9     | 74    | 70-74  | men    | Neuropsychiatric | NoEvent |

**Costs and QALYs.** `tm_add_outcomes()` gives each year its cost, intervention cost, QALY and discount factor:

```
out <- tm_add_outcomes(sim, costs, utilities, intervention, death, settings)
out[id == 1, .(arm, cycle, state, cost, intervention_cost, qaly, discount)]
```

|    | arm    | cycle | state   | cost  | intervention_cost | qaly  | discount  |
|----|--------|-------|---------|-------|-------------------|-------|-----------|
|    | <char> | <int> | <char>  | <num> | <num>             | <num> | <num>     |
| 1: | iabc   | 0     | NoEvent | 2000  | 0                 | 0.75  | 1.0000000 |
| 2: | iabc   | 1     | NoEvent | 2050  | 50                | 0.75  | 0.9708738 |
| 3: | iabc   | 2     | NoEvent | 2000  | 0                 | 0.75  | 0.9425959 |
| 4: | iabc   | 3     | NoEvent | 2000  | 0                 | 0.75  | 0.9151417 |
| 5: | iabc   | 4     | NoEvent | 2000  | 0                 | 0.75  | 0.8884870 |
| 6: | iabc   | 5     | NoEvent | 2000  | 0                 | 0.75  | 0.8626088 |
| 7: | iabc   | 6     | NoEvent | 2000  | 0                 | 0.75  | 0.8374843 |
| 8: | iabc   | 7     | NoEvent | 2000  | 0                 | 0.75  | 0.8130915 |
| 9: | iabc   | 8     | NoEvent | 2000  | 0                 | 0.75  | 0.7894092 |

|     |            |   |         |      |   |      |           |
|-----|------------|---|---------|------|---|------|-----------|
| 10: | iabc       | 9 | NoEvent | 2000 | 0 | 0.75 | 0.7664167 |
| 11: | usual_care | 0 | NoEvent | 2000 | 0 | 0.75 | 1.0000000 |
| 12: | usual_care | 1 | NoEvent | 2000 | 0 | 0.75 | 0.9708738 |
| 13: | usual_care | 2 | NoEvent | 2000 | 0 | 0.75 | 0.9425959 |
| 14: | usual_care | 3 | NoEvent | 2000 | 0 | 0.75 | 0.9151417 |
| 15: | usual_care | 4 | NoEvent | 2000 | 0 | 0.75 | 0.8884870 |
| 16: | usual_care | 5 | NoEvent | 2000 | 0 | 0.75 | 0.8626088 |
| 17: | usual_care | 6 | NoEvent | 2000 | 0 | 0.75 | 0.8374843 |
| 18: | usual_care | 7 | NoEvent | 2000 | 0 | 0.75 | 0.8130915 |
| 19: | usual_care | 8 | NoEvent | 2000 | 0 | 0.75 | 0.7894092 |
| 20: | usual_care | 9 | NoEvent | 2000 | 0 | 0.75 | 0.7664167 |

  

|  | arm    | cycle | state  | cost  | intervention_cost | qaly  | discount |
|--|--------|-------|--------|-------|-------------------|-------|----------|
|  | <char> | <int> | <char> | <num> | <num>             | <num> | <num>    |

**Summary.** `tm_summarise()` adds up each arm:

```
one_run <- tm_summarise(out, death, settings)
one_run[, .(arm, patients, cost_per_patient, qaly_per_patient, deaths)]
```

|    | arm        | patients | cost_per_patient | qaly_per_patient | deaths |
|----|------------|----------|------------------|------------------|--------|
|    | <char>     | <int>    | <num>            | <num>            | <int>  |
| 1: | iabc       | 5000     | 13123.60         | 3.593847         | 1845   |
| 2: | usual_care | 5000     | 13219.94         | 3.589423         | 1846   |

**Comparison.** `tm_compare()` takes iABC minus usual care:

```
tm_compare(one_run[, `:=`(run = 1L, seed = settings$seed)])$overall
```

|    | delta_cost | delta_intervention_cost | delta_qaly  | icer      | quadrant |
|----|------------|-------------------------|-------------|-----------|----------|
|    | <num>      | <num>                   | <num>       | <num>     | <char>   |
| 1: | -96.34138  | 37.44662                | 0.004423595 | -21778.98 | dominant |

  

|    | delta_ischaemic_stroke | delta_haemorrhagic_stroke | delta_major_bleed | delta_deaths |
|----|------------------------|---------------------------|-------------------|--------------|
|    | <num>                  | <num>                     | <num>             | <num>        |
| 1: | -33                    | -10                       | -1                | -1           |

This is exactly the first run of `tm_run()`, which uses the same seed:

```
results$by_run[run == 1, .(arm, patients, cost_per_patient, qaly_per_patient, deaths)]
```

|    | arm        | patients | cost_per_patient | qaly_per_patient | deaths |
|----|------------|----------|------------------|------------------|--------|
|    | <char>     | <int>    | <num>            | <num>            | <int>  |
| 1: | iabc       | 5000     | 13123.60         | 3.593847         | 1845   |
| 2: | usual_care | 5000     | 13219.94         | 3.589423         | 1846   |